

Serving Sentiment: Identifying Key Themes in McDonald's Customer Reviews

Introduction: It's a Friday night and you feel your stomach begin to grumble. Time for dinner! You walk over, open your fridge only to realize you have an onion, ketchup, and half a jar of pickles. This is when you decide to fall back on ol' reliable: McDonald's. You arrive, look to your left and see an 80-year old man screaming across the counter because the ice cream machine is down, again. You then look to your right and see a group of kids having the time of their life playing with the toys they acquired from their Happy Meal. You begin to wonder how two people can experience the same place, service, and food yet elicit polar opposite responses as far as their experience at McDonald's. However, walking up to each person in every McDonald's across the country, even the world, to assess what was "make or break" about their experience would be a life's work one person could not nearly accomplish on their own. You begin to wonder, how can I identify what are the most common reasons for someone having a positive experience at McDonald's and what aspects contribute to a person's negative experience the most.

Deliverables: In this case study, you will be given a data set of nearly 34,000 McDonald's restaurant reviews across the United States containing the review, rating, topic, sentiment indicator, and other necessary variables. Your goal from this data set is to perform Latent Dirichlet Allocation (LDA) to perform sentiment analysis identifying the most common themes in positive and negative McDonald's restaurant reviews to pinpoint areas most in need of improvement. The purpose of this task is to translate everyday restaurant reviews into something actionable that the McDonald's organization can use to allocate resources to make meaningful change to their customer experience and business as a whole.